



MARTÍN ARNALDO CERESA

Postdoc Researcher

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TECH LANGUAGES

Haskell Coq Lean Agda

Go C Python Rust SML Erlang

Z CSP TLA/TLA+

TECH STACK

ViM XMonad Hakyll Git L^AT_EX

Markdown GNU/Linux Emacs (Doom)

Org

LANGUAGES

Spanish: Native | English: Fluent

Portuguese: Basic

REFERENCES

Prof. César Sánchez

IMDEA Software Institute
cesar.sanchez@imdea.org

Prof. Mauro Jaskelioff

FCEIA & IOG
mjaskelioff@gmail.com

PROJECTS

Termina 2023 – Ongoing

Transpiler from Termina to C plus a RTOS runtime for embedded systems. Designed and implemented in Haskell and runtime in C.

RESEARCH EXPERIENCE

Postdoc Researcher | IMDEA Software Institute

03 2023 – Ongoing Madrid, Spain

- Ongoing research: formalizing Fraud Proof games in *Lean*.
- Studied offline monitoring properties of smart-contracts.
- Designed and implemented (in Haskell) Termina, a **non**-turing complete language for embedded software.
- Developed a simulation environment for the Tezos blockchain smart contracts in Go.
- Guided PhD students and interns.

Research Programmer | IMDEA Software Institute

10 2022 – 02 2023 Madrid, Spain

Implemented (in Rust) an intermediate layer expanding previous Tezos indexer features to monitor Tezos smart-contract with HLola.

Visiting PhD Student | IMDEA Software Institute

08 2020 – 09 2022 Madrid, Spain

- Implemented a Coq library, *Multi*, to prove and reason about multi-contract interaction inspired on the computational model of the Tezos blockchain.
- Formalized (in Coq) different execution strategies, inspired by Tezos execution model, while introducing online and onchain smart-contract monitoring capabilities.
- Participated in the development of a theory to introduce online transaction monitors to blockchain technologies.
- Participated in the implementation and development of Setchain, a new Byzantine-tolerant distributed data-structure.

Undergraduate Intern | IMDEA Software Institute

03 2013 – 09 2013 Madrid, Spain

- Expanded the strategy toolbox of EasyCrypt, a modern proof assistant for cryptographers.
- Learned about modern cryptography proofs and game simulation-based proofs.
- Implemented (in OCaml) two different tactics: field/ring equality and automatic optimistic sampling.

FraudProofs 📅 2024 – Ongoing 🌐

Mechanization of Fraud-Proof arbitration games in L2 Optimistic Rollups. Implemented in Lean4, focused on proving honest players always win.

Smart Multi Contract Interaction

📅 2021 – 2022 🌐

Designed and implemented a Coq library to reason about multi smart-contract interaction following the computational model of the Tezos Blockchain.

Haskell Lola 📅 2019 – 2020 🌐 🌐

Designed the core structure implementing HLola. Implemented the main abstraction lifting generic types of Haskell employed as data-theories in Lola.

QuickFuzz 📅 2016 – 2017 🌐

Out-of-the-box fuzzy generation of arbitrary files. Implemented Haskell arbitrary instances for arbitrary data structures in Haskell.

MegaDeTh 📅 2016 – 2017 🌐

Aggressive and generic instance derivation for recursive and mutually dependent Haskell data-types. Implemented entirely in Template Haskell.

EasyCrypt 📅 2013 🌐

Implemented tactics to simplify equations over rings and fields. Designed and implemented automatic optimistic sampling.

EDUCATION

PhD in Informatics | UNR

📅 04 2015 – 04 2023 📍 Rosario, Argentina

- Thesis: Effectful Improvement Theory.
- Summary: An improvement theory for programming languages with algebraic effects.
- Advisor: PhD. Mauro Jaskelioff

Computer Science Degree | UNR

📅 03 2008 – 03 2015 📍 Rosario, Argentina

- Title: Simulation of Parallel Programs in Haskell
- Summary: A graphic library to observe and study the “parallel structure” of parallel programs in Haskell.
- GPA: 9.60

TEACHING EXPERIENCE

Adjunct Professor | UNR

📅 2019 – 2021 📍 Rosario, Argentina

- Parallel and Concurrent Programs: I taught the basic problems and solutions of concurrent and parallel programming in C and Erlang.
- Compilers Course: I followed closely the book *Modern Compilers Implementation in ML* by Andrew Appel implementing a full-fledged compiler for the language *Tiger* without optimizations.

Teacher Assistant | UNR

📅 2011 – 2019 📍 Rosario, Argentina

My responsibilities included: assisting and conducting laboratory hours, crafting tests and final projects and grading them.

- Data Structures and Algorithms in C
- Compiler Theory and Practice
- Concurrent Programming in Erlang
- Introduction to pure programming languages and Haskell
- Parallel Thinking in Haskell
- Introduction to Lambda Calculus, STLC, Intermediate functional programming in Haskell
- Introduction to Intuitionistic Logic
- Introduction to Generalized Abstract Data Types

COURSES

λ -Calculus and Reasonable Cost Models 🌐 | UBA

📅 2019 📍 Buenos Aires, Argentina

DeepSpec Summer School 🌐 | Princeton University

📅 2018 📍 New Jersey, United States of America

Introduction to basic and algebraic Topology | UNR

📅 2018 📍 Rosario, Argentina

Haskell's Type System/GHC and its extensions 🌐 | ECI

📅 2018 📍 Río Cuarto, Argentina

FlexibleContexts, FlexibleInstances, GADTs, Phantom Types, RankN-Poly, DataKinds, MultiParamTypeClasses, FunctionalDependencies, TypeFamilies.

Introduction to Category Theory and Agda 🌐 | UNR

📅 2016 📍 Rosario, Argentina

Interactive Theorem Proving 🌐 | UBA

📅 2014 📍 Buenos Aires, Argentina

Introduction to Coq. Natural deduction, Curry-Howard correspondence and Lambda Calculus.

Introduction to Type Theory in Agda 🌐 | UNR

📅 2011 📍 Rosario, Argentina

PUBLICATIONS

Journal Articles

- M. Capretto, **M. Ceresa**, A. Fernández Anta, A. Russo, and C. Sánchez, “Improving blockchain scalability with the setchain data-type,” *Distrib. Ledger Technol.*, Oct. 2023, Just Accepted. DOI: 10.1145/3626963.
 - **M. Ceresa** and M. Jaskelioff, “Effectful improvement theory,” *Sci. Comput. Program.*, vol. 217, p. 102 792, 2022. DOI: 10.1016/j.scico.2022.102792.
 - G. Grieco, **M. Ceresa**, A. Mista, and P. Buiras, “Quickfuzz testing for fun and profit,” *Journal of Systems and Software*, vol. 134, pp. 340–354, 2017. DOI: 10.1016/j.jss.2017.09.018.
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Conference Proceedings

- M. Capretto, **M. Ceresa**, and C. Sánchez, “Monitoring the future of smart contracts,” in *Fundamental Approaches to Software Engineering - 27th International Conference, FASE 2024, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2024, Luxembourg City, Luxembourg, April 6-11, 2024, Proceedings*, D. Beyer and A. Cavalcanti, Eds., ser. Lecture Notes in Computer Science, vol. 14573, Springer, 2024, pp. 122–142. DOI: 10.1007/978-3-031-57259-3_6.
- M. Capretto, **M. Ceresa**, A. F. Anta, A. Russo, and C. Sánchez, “Setchain: Improving blockchain scalability with byzantine distributed sets and barriers,” in *IEEE International Conference on Blockchain, Blockchain 2022, Espoo, Finland, August 22-25, 2022*, IEEE, 2022, pp. 87–96. DOI: 10.1109/BLOCKCHAIN55522.2022.00022.
- M. Capretto, **M. Ceresa**, and C. Sanchez, “Transaction monitoring of smart contracts,” in *Runtime Verification*, T. Dang and V. Stolz, Eds., Cham: Springer International Publishing, 2022, pp. 162–180, ISBN: 978-3-031-17196-3.
- **M. Ceresa** and C. Sánchez, “Multi: A Formal Playground for Multi-Smart Contract Interaction,” in *4th International Workshop on Formal Methods for Blockchains (FMBC 2022)*, Z. Dargaye and C. Schneidewind, Eds., ser. Open Access Series in Informatics (OASlcs), vol. 105, Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022, 5:1–5:16, ISBN: 978-3-95977-250-1. DOI: 10.4230/OASlcs.FMBC.2022.5.
- **M. Ceresa**, F. Gorostiaga, and C. Sánchez, “Declarative stream runtime verification (hlola),” in *Programming Languages and Systems*, B. C. d. S. Oliveira, Ed., Cham: Springer International Publishing, 2020, pp. 25–43, ISBN: 978-3-030-64437-6.
- G. Grieco, **M. Ceresa**, and P. Buiras, “Quickfuzz: An automatic random fuzzer for common file formats,” in *Proceedings of the 9th International Symposium on Haskell, Haskell 2016, Nara, Japan, September 22-23, 2016*, G. Mainland, Ed., ACM, 2016, pp. 13–20. DOI: 10.1145/2976002.2976017.